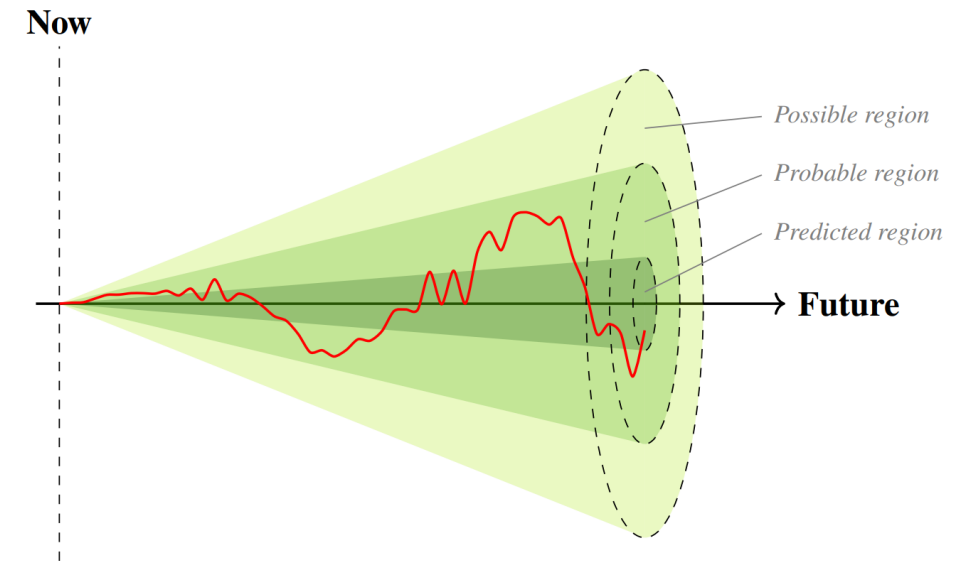


I/O-pipeline for Scenario-Based Transport Planning under Deep Uncertainty

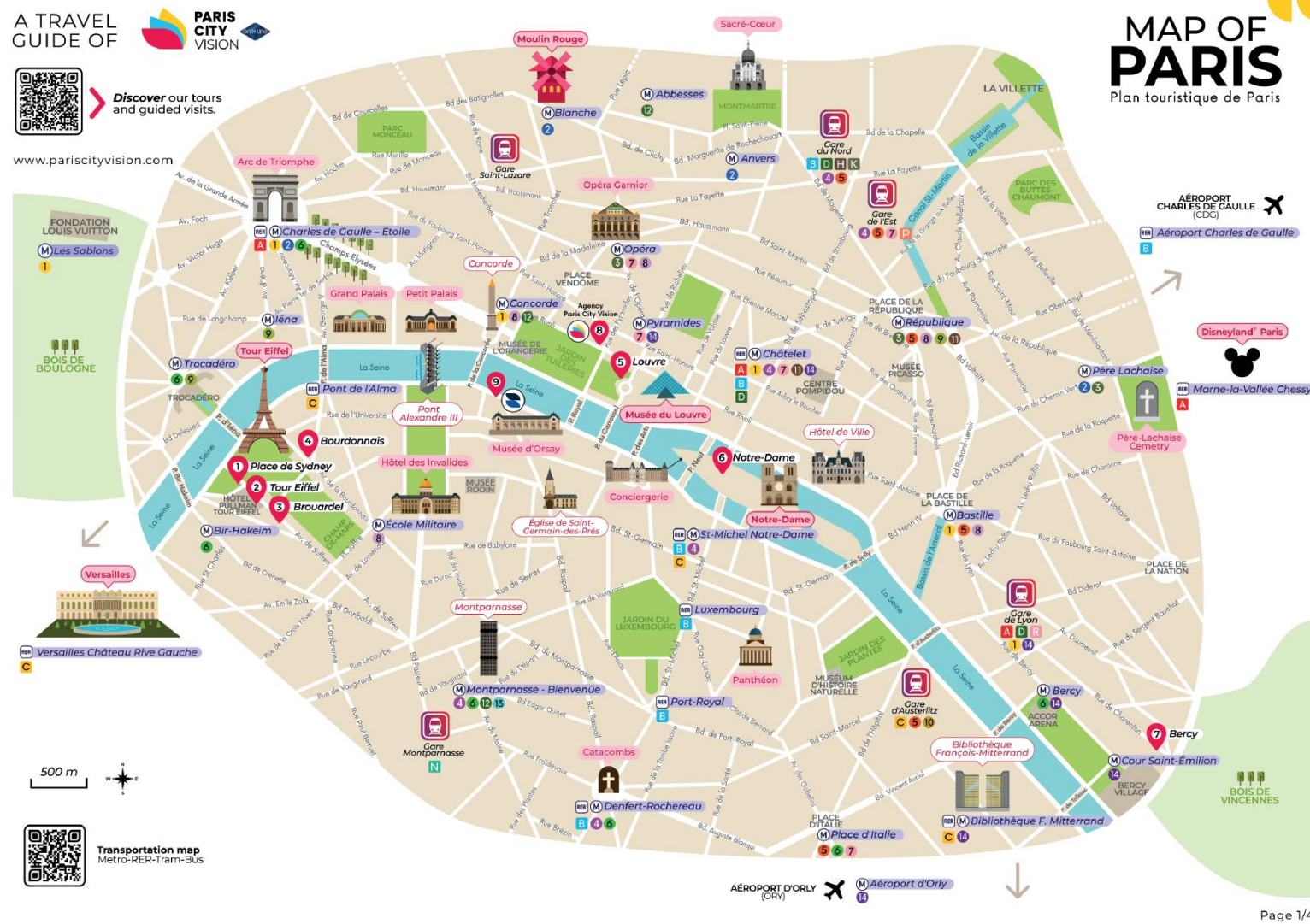
MuM Paris 2026

Jan Andrin Ghadamian¹, Orlando Roman¹, Milos Balac², Bryan Adey¹

ETH Zürich,
1: D-BAUG, IBI-IM
2: D-MAVT, CSFM



Motivation



Motivation



Motivation

1967

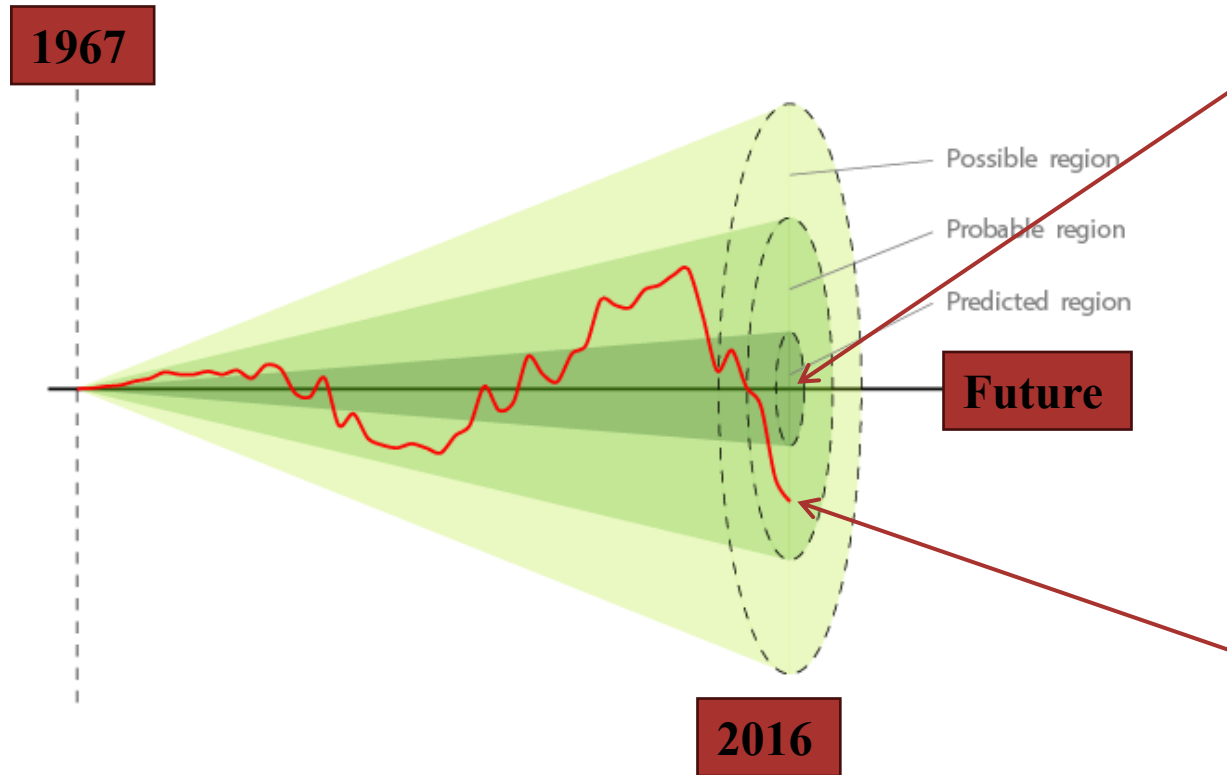


Motivation

2016



Motivation



Motivation

“Transport modelling is an attempt to predict something that will inevitably be wrong.”

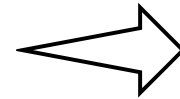
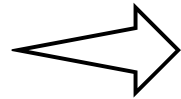
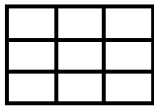
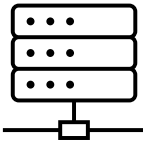
Nevertheless, for long-term planning, the value of transport models lies not in their predictive accuracy, but rather in their ability to systematically explore plausible futures and support evidence-based decision-making.



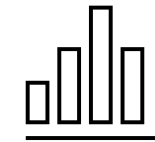
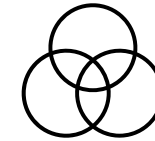
I/O-pipeline

Input

- GTFS
- OSM
- census
- ...



Output

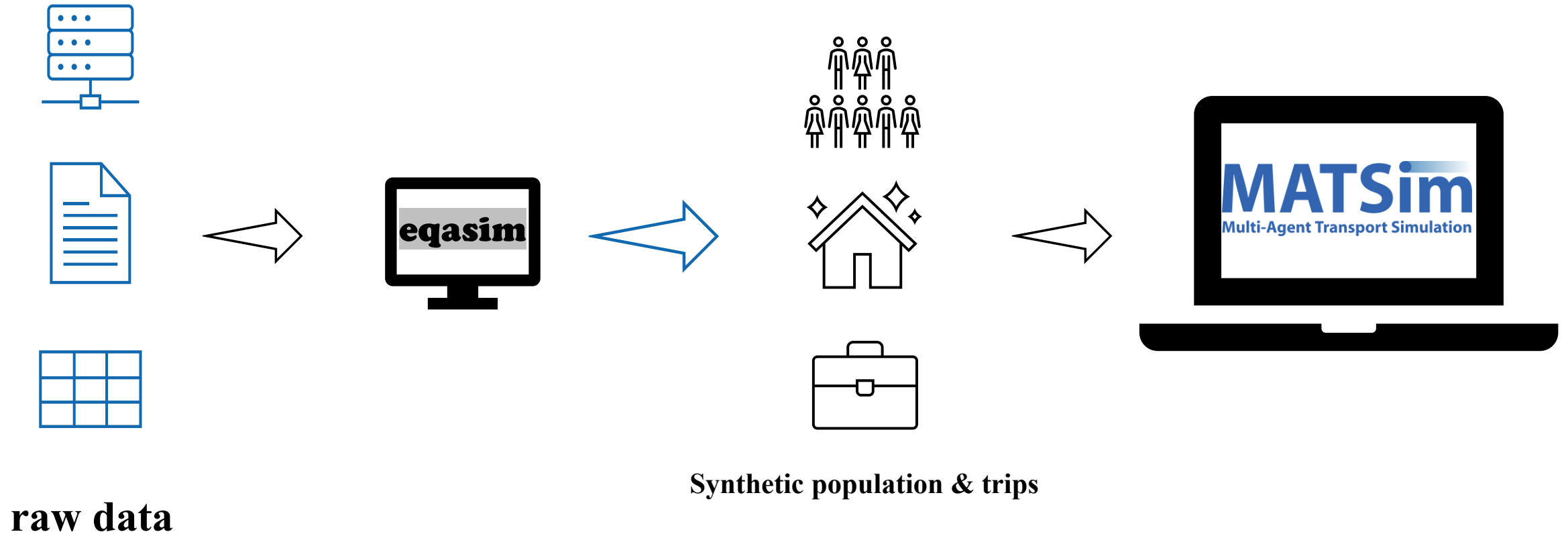


- Pkm
- ph
- Modal split
- ...



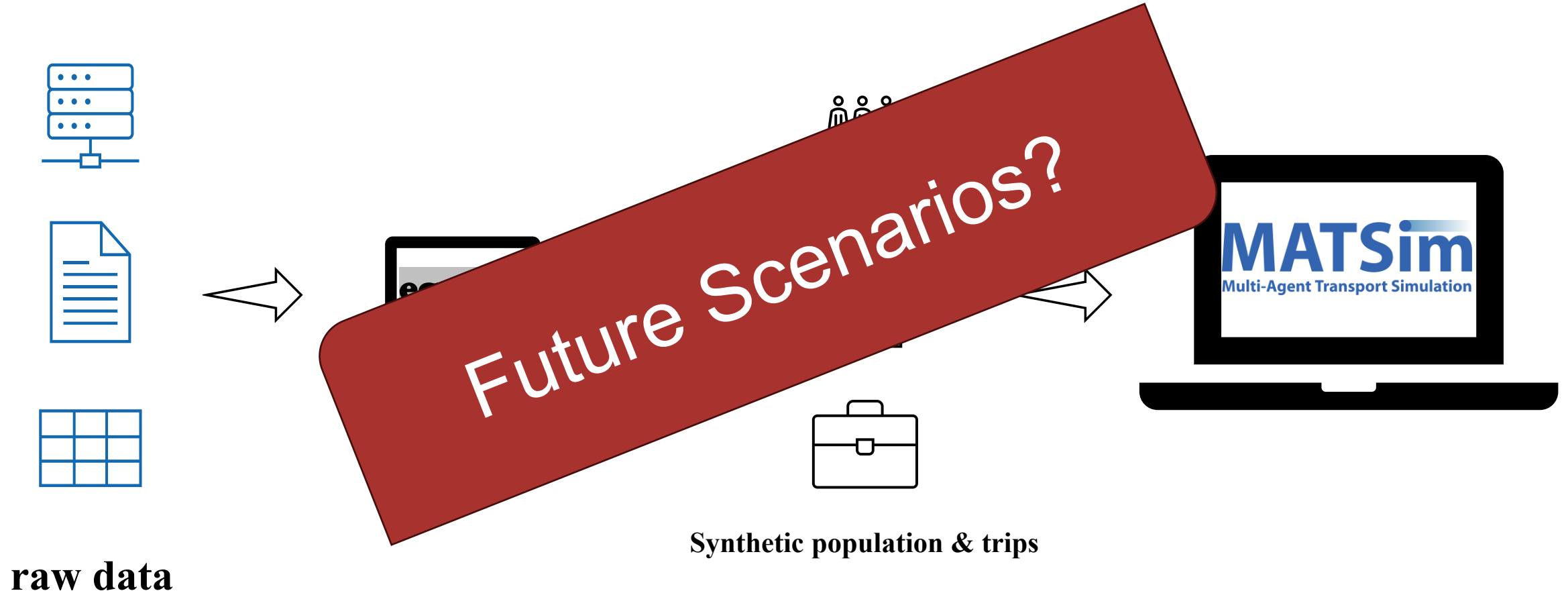
I/O-pipeline

Model of calibrated current-state representation

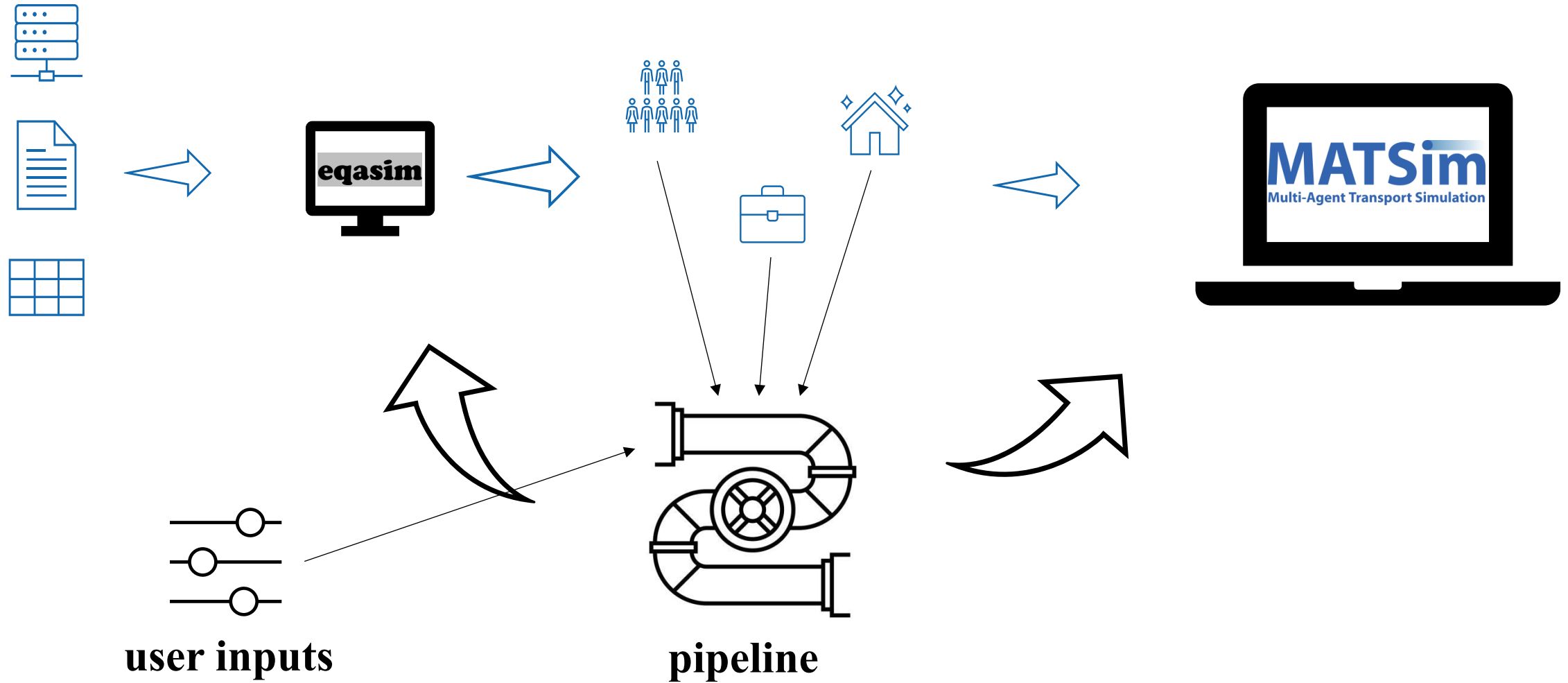


I/O-pipeline

Model of calibrated current-state representation

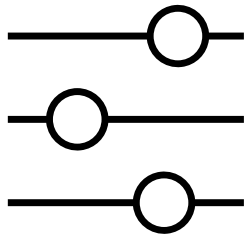


I/O-pipeline

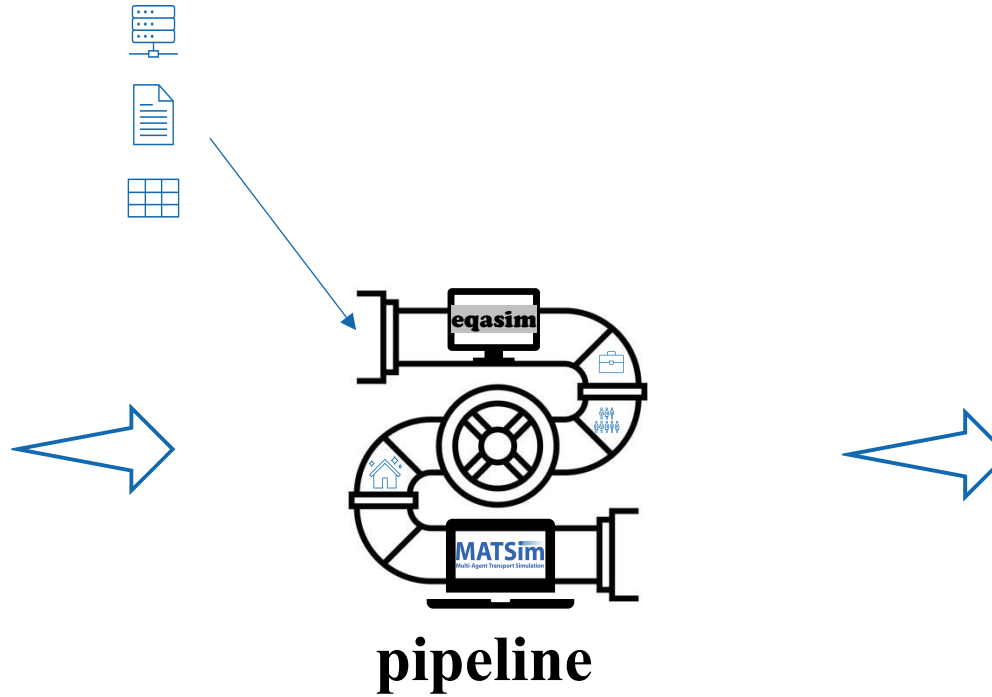


I/O-pipeline

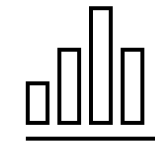
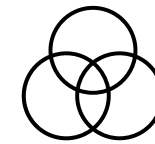
Input



- Population size
- mode preference
- Network changes
- ...



Output

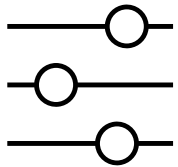


- Pkm
- ph
- Modal split
- ...



I/O-pipeline

Input

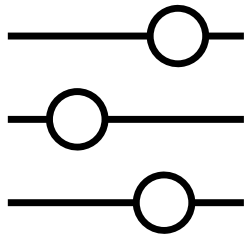


Level	Driver	Integration Approach
A – Direct	Economy, Behaviour, Living, Population, Labour	Configuration-level adjustments to cost functions, utility parameters, and synthetic population or activity attributes
B – Moderate	Technology, Wealth, Journeys, Leisure Time	Dedicated MATSim extensions or extended trip generation logic; requires careful validation
C – Complex	Regulations, Energy, Goods	Multi-dimensional; requires heterogeneous data sources and coordinated implementation across pipeline stages
D – Unfeasible	Climate, Security	Outside the scope of daily activity-based simulation; incompatible timescales or singular disruptive nature

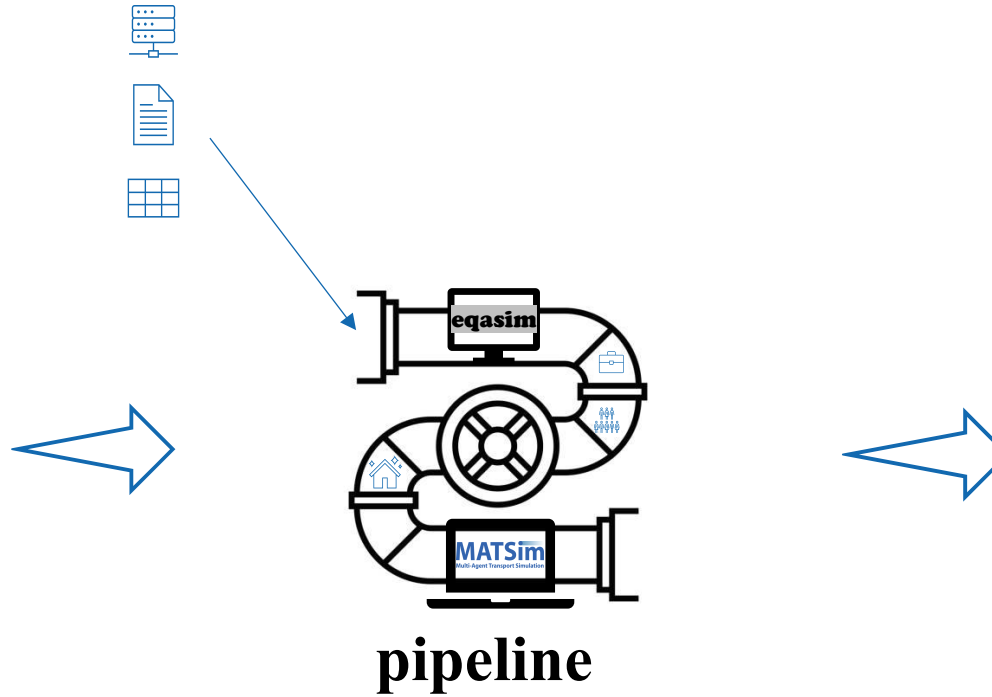


I/O-pipeline

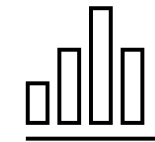
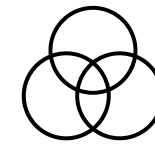
Input



- Population size
- mode preference
- Network changes
- ...



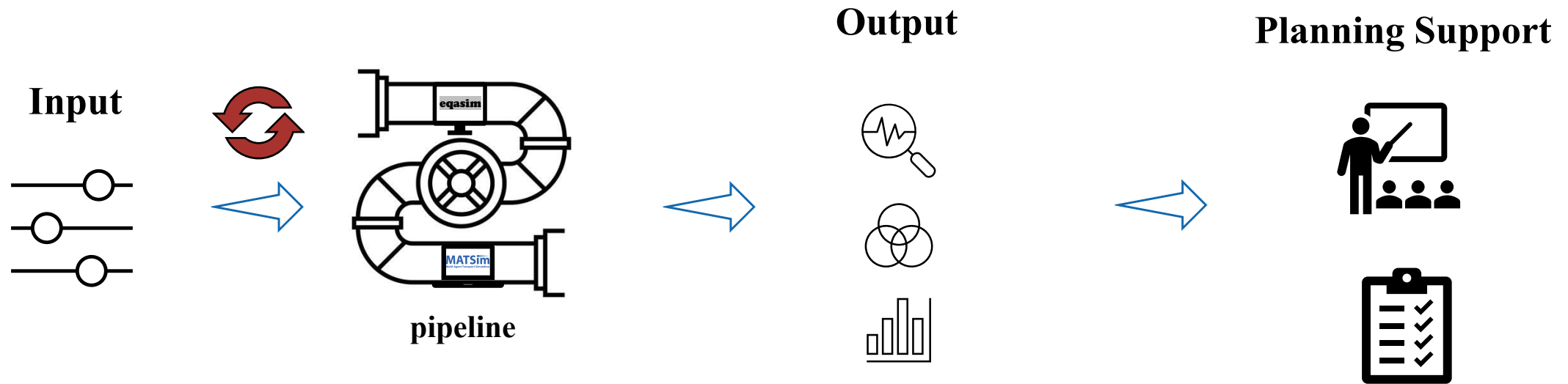
Output



- Pkm
- ph
- Modal split
- ...



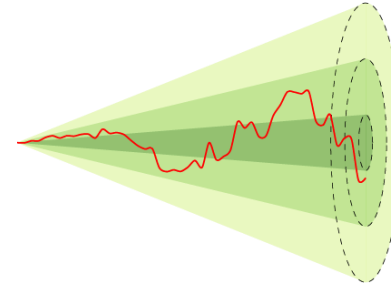
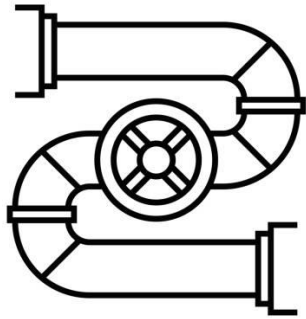
I/O-pipeline



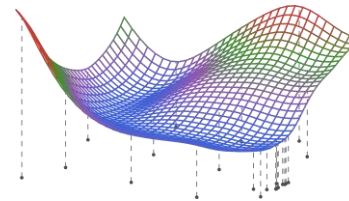
Input-to-Output pipeline
(I/O-pipeline)



I/O-pipeline - Goal



- Individual scenario exploration

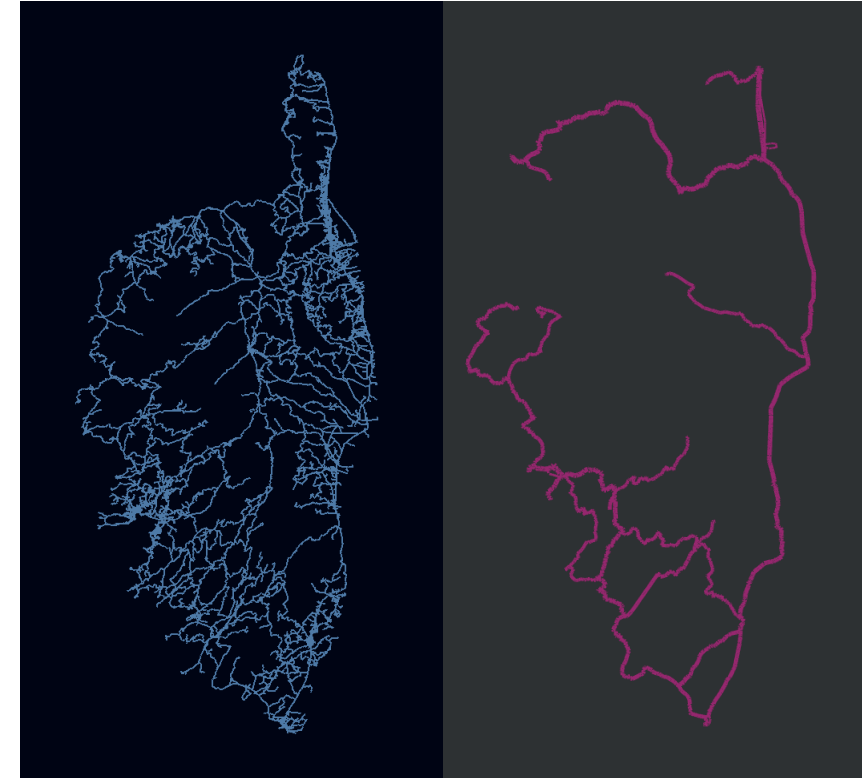


- systematic scenario generation for surrogate modelling



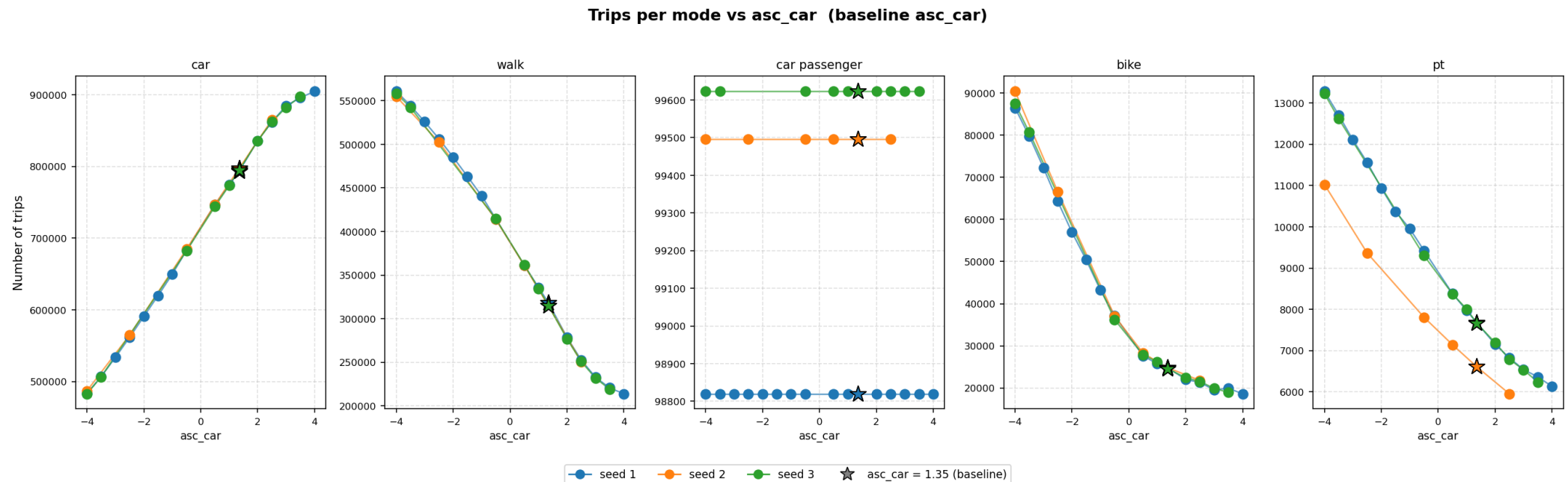
Results - Setting

- **Testbed Corsica**
 - Size: 350'000 inhabitants (agents)
 - Island: no external influences
 - Data availability
 - Eqasim documentation
- **Three types of output**
 1. Sensitivity Analysis
 2. Latin Hyper Cube Sampling – Surrogate Modelling
 3. Graphical User Interface



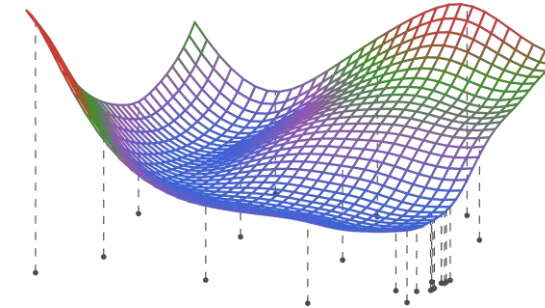
Results - 1

- Sensitivity Analysis on user preference and population parameters:
 - ASC walking $[-4, 4]$
 - ASC car $[-4, 4]$
 - ASC bike $[-4, 4]$
 - Population growth factor $[0.75, 1.25]$



Results - 2

- Latin Hyper Cube Sampling - Surrogate output:
 - **First stage:** generalised output that can/could be used to calculate the modal split.
 - E.g. total kilometers travelled per mode, total trips per mode, total person hours travelled per mode.
 - **Second stage:** same outputs but spatially explicit. Either for pre-defined zones or smaller «artificial» zones.
 - E.g. kilometers travelled per mode in zone x , total trips per mode in zone y .
 - **Third stage:** explicitly spatial distribution. Flow on each link (either peak hour/max value) or daily avg. Or both...
 - E.g. total trips travelled per link and mode.



Results - 2

3.2 trips_walk

Normalized (modified) leave-one-out error: 5.013×10^{-3} .

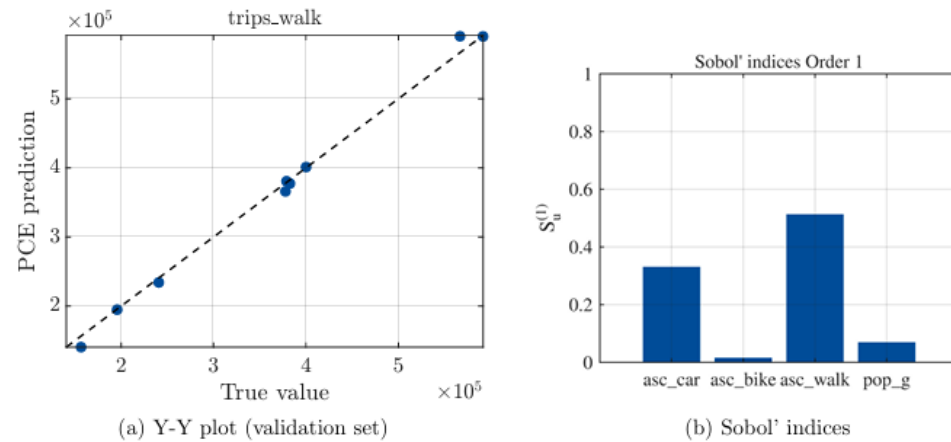
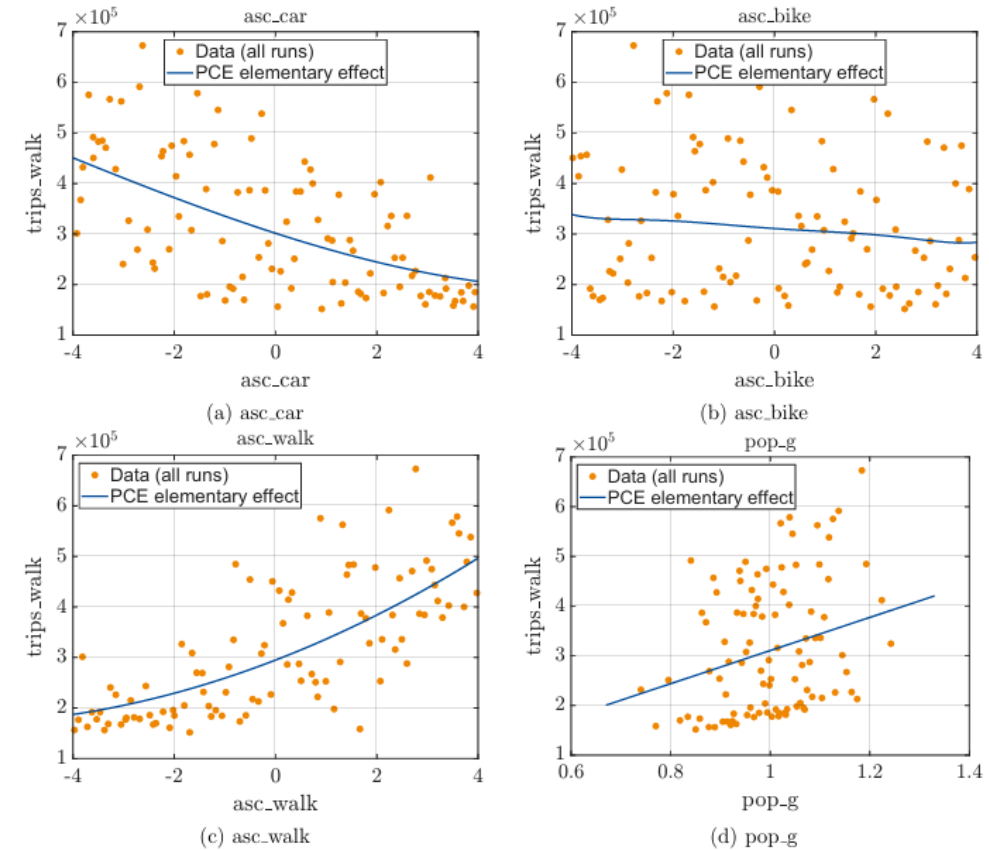


Figure 3: Results for output `trips_walk`

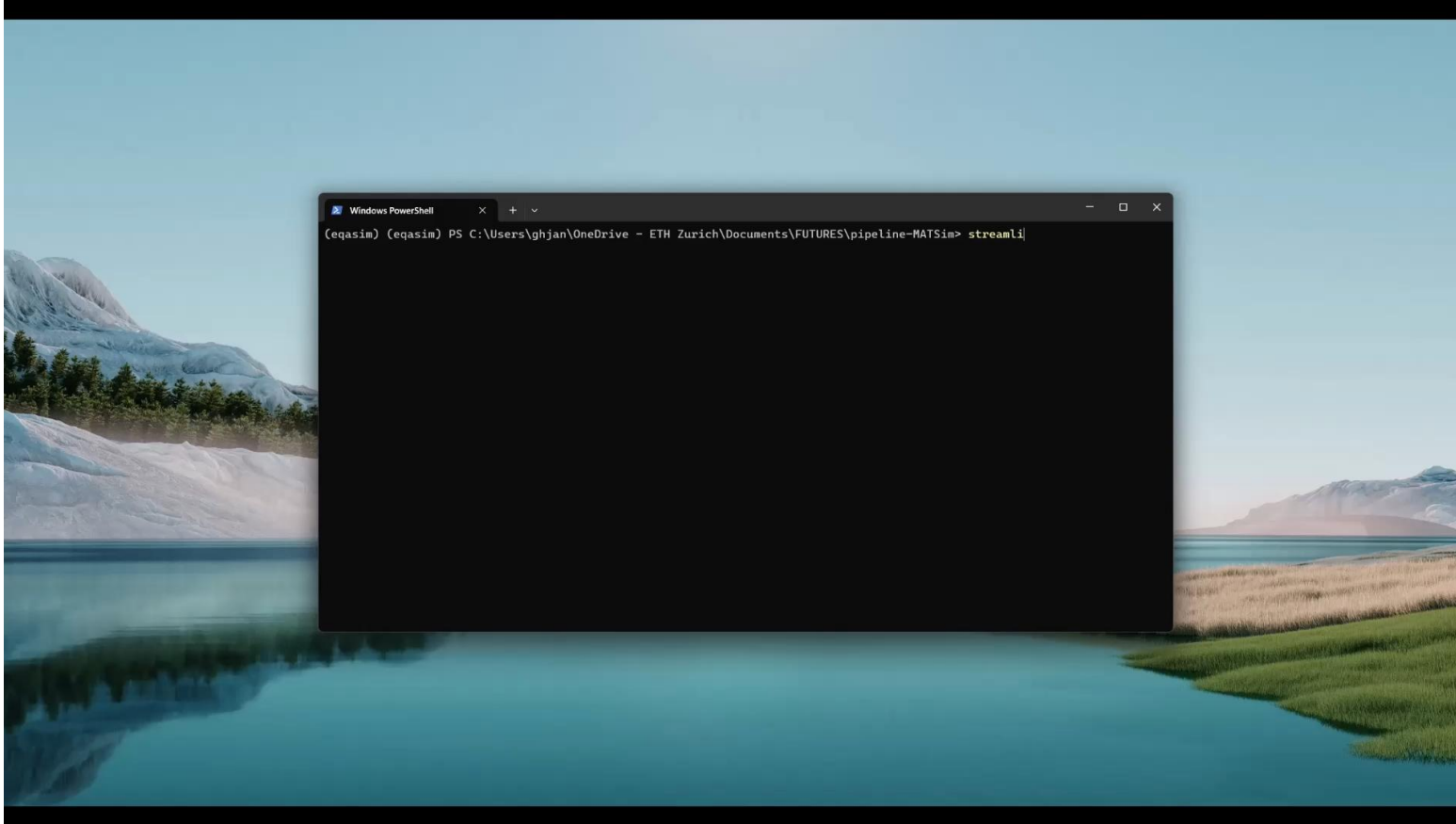
Table 3: Sobol' indices for output `trips_walk`

Input	First order	Total
asc_car	0.3313	0.3918
asc_bike	0.0160	0.0226
asc_walk	0.5133	0.5753
pop_g	0.0698	0.0809



Results - 3

- Graphical User Interface



Thank you for your attention!

Questions?



Appendix

- References:
 - Paris Map (Slides 2,3): www.pariscityvision.com
 - Image “Voie George Pompidou” (Slides 4,5,6): Facebook IMM Design Lab
 - Surrogate Model (Slide 7): Presentation on Uncertainty by Orlando Roman



Results - 2

